Computational Statistics –the emergence of computational statistics was stimulated by the availability of very large data sets. These data are often generated by heterogeneous mechanisms, from unknown process, or from non-random mechanism. The availability of powerful computing facilities and efficient algorithms simplified the analysis of these data. Oftentimes, the methods used are iterative in nature, thus, exact sampling distributions are not mathematically tractable. Computational statistics is a venue where statistical theory evolves from a dominantly nonparametric framework and uses computing technology.

STAT 210 STATISTICAL SOFTWARE.
Database management and programming using statistical software.

STAT 211 STATISTICAL COMPUTING.
Algorithms for statistical computing; numerical analysis for linear and nonlinear models; random number generation; Monte Carlo methods.

STAT 226 APPLIED MULTIVARIATE ANALYSIS.
Multivariate normal distribution; principal components analysis; biplots and h-plots; factor analysis; discriminant analysis; cluster analysis; multidimensional scaling; correspondence analysis; canonical correlation analysis; graphical and data oriented techniques; applications
Co-requisite: Stat 223 or its equivalent

STAT 240 HIGH DIMENSIONAL DATA .
High dimensional data; high dimensional data visualization; high dimensional data analysis; dimension reduction; pattern search; clustering; applications
Pre-requisite: Stat 226/Equivalent, Stat 223/Equivalent

STAT 247 DATA MINING AND BUSINESS INTELLIGENCE.
Principles of data mining; methods of data mining; themes of data mining; applications of data mining in business intelligence.
Pre-requisite: Pre-requisite: Stat 210/Equivalent, Stat 226/Equivalent,
Co-requisite: Stat 223/Equivalent, and Stat 225/Equivalent

STAT 249 NONPARAMETRIC MODELING.
Smoothing methods; kernel smoothing; spline smoothing; regression trees; projection pursuit; nonparametric regression; cross-validation; scoring; high dimensional predictors; additive models; backfitting.
Pre-requisite: Stat 222/232 and Stat 223/Equivalent

STAT 252 BOOTSTRAP METHODS.
Empirical distribution functions; resampling and nonparametric statistical inference; optimality of the bootstrap; bootstrap in hypothesis testing; bootstrap in confidence intervals; bootstrap in regression models; bootstrap for dependent data.
Pre-requisite: Stat 222/232 and Stat 223/Equivalent

STAT 262 NONPARAMETRIC STATISTICS.
Distribution-free statistics; U-statistics; power functions; asymptotic relative efficiency of tests; confidence intervals and bounds; point estimation; linear rank statistics; other methods for constructing distribution-free procedures.
Pre-requisite: Stat 232

STAT 265 ROBUST STATISTICS.
Breakdown point and robust estimators; M-, R-, and L-estimates; robust tests; robust regression and outlier detection.
Pre-requisite: Stat 232

STAT 267 ADVANCED APPLIED MULTIVARIATE ANALYSIS.
Confirmatory factor analysis; multidimensional scaling; correspondence analysis; classification trees; CHAID; procrustes analysis; neural networks; structural equation modeling.
Pre-requisite: Stat 226 or equivalent/COI

STAT 277 STATISTICS FOR IMAG E AN ALYS IS.
Radiometric enhancement techniques; geometric enhancement using image domain techniques; multispectral transformation of data; supervised classification techniques; clustering and unsupervised classification; applications.
Pre-requisite: Consent of Instructor

Required Classes

Stat 100 Calculus and Matrix Algebra for Statistics
Stat 195 Introduction to Mathematical Statistics
Stat 221 Introductory Probability
Stat 251 Survey Operations
Stat 222 Introduction to Statistical Inference
Stat 223 Applied Regression Analysis
Stat 250 Sampling Designs
Stat 290 Statistical Consulting
Stat 298 Special Problem

Statistical Methods for Market Research and Business Intelligence - from data collected through face-to-face interviews, focused group discussions, product testing, etc., the paradigm of market research is gradually changing. Data is now generated from actual usage/purchase history of consumers, cases where streaming data is generated from continuous consumer activities. While the data collection does not really follow some probabilistic methods, their volume is huge enough to treat it as approximating the population. This area incorporates both the 'traditional' and modern methods of market research leading towards business intelligence.

STAT 210 STATISTICAL SOFTWARE.
Database management and programming using statistical software.

STAT 211 STATISTICAL COMPUTING.
Algorithms for statistical computing; numerical analysis for linear and nonlinear models; random number generation; Monte Carlo methods.

STAT 225 TIME SERIES ANALYSIS.
Classical procedures; stationarity; Box-Jerkins modeling procedure: autocorrelation function, partial autocorrelation function; identification, estimation, diagnostic checking, forecasting; transfer functions; applications.
Pre-requisite: Stat 223 or equivalent

STAT 226 APPLIED MULTIVARIATE ANALYSIS.
Multivariate normal distribution; principal components analysis; biplots and h-plots; factor analysis; discriminant analysis; cluster analysis; multidimensional scaling; correspondence analysis; canonical correlation analysis; graphical and data oriented techniques; applications
Co-requisite: Stat 223 or its equivalent

STAT 241 NONLINEAR REGRESSION.
Classification of nonlinear models; iterative estimation and linear approximation; practical considerations: model specification, starting values, transformations; convergence; multiresponse model; models from differential equations; nonlinear inference regions; measures of nonlinearity; applications.
Pre-requisite: Stat 223 or equivalent

STAT 243 CATEGORICAL DATA ANALYSIS.
Cross-classified tables, multidimensional tables; loglinear model; logit models, measures of association; inference for categorical data; applications
Pre-requisite: Stat 223 or equivalent

STAT 247 DATA MINING AND BUSINESS INTELLIGENCE.
Principles of data mining; methods of data mining; themes of data mining; applications of data mining in business intelligence.
Pre-requisite: Pre-requisite: Stat 210/Equivalent, Stat 226/Equivalent,
Co-requisite: Stat 223/Equivalent, and Stat 225/Equivalent

STAT 266 APPLIED NONPARAMETRIC METHODS.
Methods for single, two and k samples; trends and association; nonparametric bootstrap.
Pre-requisite: Stat 222 and Stat 223

STAT 267 ADVANCED APPLIED MULTIVARIATE ANALYSIS.
Confirmatory factor analysis; multidimensional scaling; correspondence analysis; classification trees; CHAID; procrustes analysis; neural networks; structural equation modeling.
Pre-requisite: Stat 226 or equivalent/COI

STAT 268 ADVANCED TIME SERIES ANALYSIS.
Nonstationarity; cointegration; interventions models; state space models; transfer functions; frequency domain; panel data; nonparametric methods for time series; nonparametric prediction; AR-Sieve; block bootstrap
Pre-requisite: Stat 223/Equivalent and 225/Equivalent

STAT 270 EXPLORATORY DATA ANALYSIS.
Graphical methods; single batch analysis and analysis of several batches; order statistics; resistant estimators; robust tests; robust regression; median polish; applications.
Pre-requisite: Stat 222 or equivalent

STAT 274 MARKET RESEARCH.
The marketing research; data and data generation in marketing research; analytical methods; consumer behavior modeling.
Pre-requisite: Stat 223/Equivalent and Stat 226/Equivalent

STAT 276 STATISTICS FOR GEOGRAPHIC INFORMATION SYSTEMS.
Components of a geographical information system, data structures and elements of spatial modeling; exploratory spatial data analysis; quadrat analysis, tesselations and spatial autocorrelation; spatial modeling and prediction; some sampling theory; applications.
Pre-requisite: Consent of Instructor